Addressing challenge 1:

$$J^{PPO}(\theta) = E_t\left[\min\left(\frac{\pi_\theta(a_t|s_t)}{\pi_{old}(a_t|s_t)}A_t, clip\left(\frac{\pi_\theta(a_t|s_t)}{\pi_{old}(a_t|s_t)}, 1-\varepsilon, 1+\varepsilon\right)A_t\right)\right]$$
$$= E_t\left[\frac{1}{\pi_{old}(a_t|s_t)}\min\left(\pi_\theta(a_t|s_t)A_t, clip\left(\pi_\theta(a_t|s_t), (1-\varepsilon)\pi_{old}(a_t|s_t), (1+\varepsilon)\pi_{old}(a_t|s_t)\right)A_t\right)\right]$$

Replace π_{old} with π_{behav}

Replace π_{old} with π_{prox}

Replace π_{old} with π_{prox}

$$J^{decoupled-PPO}(\theta) = E_t\left[\frac{1}{\pi_{behav}(a_t|s_t)}\min\left(\pi_\theta(a_t|s_t)A_t, clip\left(\pi_\theta(a_t|s_t), (1-\varepsilon)\pi_{prox}(a_t|s_t), (1+\varepsilon)\pi_{prox}(a_t|s_t)\right)A_t\right)\right]$$
$$= E_t\left[\min\left(\frac{\pi_\theta(a_t|s_t)}{\pi_{behav}(a_t|s_t)}A_t, \frac{\pi_{prox}(a_t|s_t)}{\pi_{behav}(a_t|s_t)}clip\left(\frac{\pi_\theta(a_t|s_t)}{\pi_{prox}(a_t|s_t)}, 1-\varepsilon, 1+\varepsilon\right)A_t\right)\right]$$

Addressing challenge 2:

Assume a response consists of T_N tokens continuously generated by N policy versions. There exists a policy π_{behav} such that the response is sampled entirely from π_{behav}:

$$\pi_{behav}(a_t|s_t) = \begin{cases} \pi_0(a_t|s_t) & 0 < t \le T_1 \\ \pi_1(a_t|s_t) & T_1 < t \le T_2 \\ \pi_2(a_t|s_t) & T_2 < t \le T_3 \\ \cdots & \\ \pi_{N-1}(a_t|s_t) & T_{N-1} < t \le T_N \end{cases}$$

Proof:
Given a query q, an autoregressive LLM generates a sequence $(q, a_1, a_2, a_3, \ldots)$. Let S_t denotes all possible sequences whose length is t (exclude q). For instance, $S_1 = (q, a_1), S_2 = (q, a_1, a_2), S_3 = (q, a_1, a_2, a_3)$. It is obviously that $S_i \cap S_j = \emptyset$ for $i \neq j$.